Mwagonubah

THE MALAWI NATIONAL EXAMINATIONS BOARD

2003 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

AGRICULTURE

Subject Number: M012/II

Wednesday, 5 November

Time Allowed: 2 hour sessions
8:30 am onwards

PAPER II

(100 marks)

Practical

Instructions

1. This paper contains 4 pages. Please check.

2. Answer all questions. Use the Answer Book provided. Each question carries 20 marks. The marks are indicated against each section of the question.

3. Write your **Examination Number** on all pages of your Answer Book.

4. At the end of the examination, hand in your answer book to the invigilator.

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1. You are provided with samples of livestock feed labelled **X, Y** and **Z**.

a. Identify the feed samples labelled **X, Y** and **Z**. **(3 marks)**

b. (i) Explain **two** reasons for giving sample **Y** to layers. **(4 marks)**

(ii) Explain any **two** reasons for providing green vegetables to layers besides taking sample **Y**. **(4 marks)**

(iii) State any **two** reasons for providing a small amount of sample **X** to layers. **(2 marks)**

c. (i) Explain **one** problem a chicken feeding only on sample **Z** would eventually have. **(2 marks)**

(ii) How would you correct the problem in **Question c(i)**? **(1 mark)**

d. If you were to prepare a simple ration for chickens from maize, madeya and beans, describe how this could be done. **(4 marks)**

2. You are provided with samples of **two** root systems.

a. Identify the root systems labelled **Q** and **R**. **(2 marks)**

b. (i) Draw the root system **Q** and label **three** parts. **(5 marks)**

(ii) Draw the root system **R** and label **two** parts. **(3 marks)**

c. Explain how the rooting depth of **Q** affects:

(i) amount of irrigation **(2 marks)**

(ii) frequency of irrigation **(2 marks)**

d. Why is the root system labelled **R** important in soil conservation? **(2 marks)**

e. Explain any **two** advantages of rotating crops with the two root systems identified in **Question 2(a)**. **(4 marks)**

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3. The **table below** shows results from Soil Science Laboratory and data from meteorological observations of an Extension Planning Area. Use it to answer the questions that follow:-

Results from Soil Science Laboratory and Data from Meteorological Station

CHARACTERISTIC	SOIL A	SOIL B	SOIL C
Percentage sand	40	42	20
Percentage silt	8	6	10
Percentage clay	50	47	60
Soil colour	red	brown	black
Effective depth (metres)	4	1	1.5
Percentage organic matter	2	5	10
Rainfall (mm per year)	2 000	1 500	1 000
Annual rainfall distribution	more even distribution	uneven distribution	uneven distribution
Temperature	medium	medium	high
Soil pH	5.1	6.2	7.4

a. (i) Which soil is likely to have the highest water holding capacity? (1 mark)

(ii) Explain **two** reasons for the answer to **Question a(i).** (4 marks)

b. (i) Which soil would be good for tree crops? (1 mark)

(ii) Explain **two** reasons for the answer to **Question b(i).** (4 marks)

c. Explain any **two** possible reasons for the acidity in **Soil A**. (4 marks)

d. (i) From which soil would the results show the highest population count of micro-organisms? (1 mark)

(ii) Explain **one** reason for the answer to **Question d(i).** (2 marks)

e. State any **three** ways of improving **Soil A.** (3 marks)

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4. Form Four students would like to find out the effect of four different crop spacings (30 cm, 60 cm, 90 cm and 120 cm) on the yield of maize in their school garden.

a. Design an experiment in **three** blocks using a randomized block design. **(7 marks)**

b. Describe how the experiment would be conducted using the following guidelines:

(i) carrying out the experiment **(3 marks)**

(ii) collection of data **(3 marks)**

(iii) recording of data **(2 marks)**

(iv) analysis of data **(2 marks)**

c. Give **one** reason for repeating an experiment. **(1 mark)**

d. In your opinion, which of the above spacings would you say is a control? Give a reason for your answer. **(2 marks)**

5. A farmer has 2 hectares of land on which he/she grows MH18. He/she however plans to make the following changes:

- To apply 4 bags of urea instead of 6 bags per hectare at K1 300.00 per bag.
- To sell 40 bags of maize at K1 000.00 per bag to Chibuku Products instead of K850.00 per bag to ADMARC.
- To store maize in 50 sacks at K30.00 each instead of storing it in the nkhokwe.
- To spend K500.00 instead of K300.00 on actellic.
- To spend K600.00/hectare instead of K300.00/hectare on casual labour.

a. Prepare a partial budget for this farmer. **(11 marks)**

b. Should this farmer go ahead with his/her plan? Explain your answer. **(3 marks)**

c. Explain **two** main uses of the partial budget prepared in **Question 5a** to this farmer. **(4 marks)**

d. Explain the major weakness of the partial budget prepared in **Question 5a.** **(2 marks)**

END OF QUESTION PAPER

NB: This paper contains 4 pages.

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